The 8-Dimensional Sovereign Profile: Multi-Dimensional Civic Identity for Post-State Governance

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Abstract

We present the 8-Dimensional Sovereign Profile, a governance credential system that replaces single-axis identity verification (Proof of Humanity, Worldcoin, Gitcoin Passport) with eight physically-grounded dimensions measured from independent on-chain source clusters. Each dimension—epistemic integrity, thermodynamic yield, network resilience, economic velocity, civic participation, stewardship, semantic resonance, and domain competence—is measured directly from verifiable activity rather than self-reports or financial proxies. Constitutional invariants enforced at the DHT validation layer (no single dimension exceeding 50% weight, mandatory exponential decay with bounded λ, AI agents capped at Steward tier) prevent dimensional capture and permanent oligarchy. Privacy is architecturally enforced through zero-knowledge STARK proofs ($\sim$1ms prove, $\sim$0.3ms verify, 7.5KB proof) that demonstrate threshold compliance without revealing raw dimensions. Deployed across 141+ Holochain zomes in 16 cluster DNAs with 87 unit tests, the system has been validated through 300-year governance simulations showing zero Guardian vetoes attempted (deterrence effect) and 100% community survival across all seeds. The system is open-source (AGPL-3.0) and published as a standalone Rust crate with zero HDK dependency.

Keywords: Decentralized Identity, Multi-Dimensional Credentials, Sybil Resistance, Zero-Knowledge Proofs, Governance, Holochain

1 Introduction

Decentralized governance systems face a fundamental credential problem: how do you determine who should participate in governance decisions, and with what weight, without reproducing the failure modes of capital-weighted voting (plutocracy) or uniform voting (ochlocracy)?

Existing approaches are single-axis solutions. **Proof of Humanity** [Kleros, 2021] uses video attestation to verify biological uniqueness but measures nothing about governance engagement. **Worldcoin** [Worldcoin, 2023] uses iris biometrics for global uniqueness but provides no information about trustworthiness or domain expertise. **Gitcoin Passport** [Gitcoin, 2023] aggregates identity signals (ENS names, social accounts, guild memberships) but remains fundamentally identity-focused, not governance-focused. **Soulbound Tokens** [Weyl et al., 2022] propose non-transferable attestations but lack temporal dynamics—a token issued in 2020 carries the same weight as one issued today.

All of these share a common structural limitation: they verify *identity* without verifying *consciousness of the governed domain*. A verified human with no engagement, no reputation, no community trust, and no domain knowledge is a verified human who can still be paid to vote, bribed to attest, or coerced to participate.

We propose a fundamentally different approach: the **8-Dimensional Sovereign Profile**, a governance credential computed from

eight independent, physically-grounded dimensions, each measured from a primary on-chain source cluster. The credential produces a continuous score in $[0, 1]$ that maps to five progressive governance tiers, decays exponentially without engagement, and is provable via zero-knowledge proofs without revealing raw dimensions.

The key architectural insight is that *dimensional independence* is the foundation of Sybil resistance. Gaming one dimension is straightforward. Gaming eight independent dimensions simultaneously—each measured from a different physical source—approaches the cost of genuine engagement.

1.1 Contributions

1. **8D credential architecture**: Eight physically-grounded dimensions with independent source clusters, replacing abstract composite scores.
2. **Constitutional envelope**: Hardcoded invariants (decay bounds, weight caps, AI tier limits) enforced at DHT validation layer—below any governance vote.
3. **Privacy-preserving proofs**: STARK circuits proving threshold compliance without revealing individual dimensions (∼1ms, 7.5KB).
4. **Hyperdimensional encoding**: 16,384-bit binary hypervector encoding of profiles for efficient similarity search and tier derivation.
5. **Empirical validation**: 300-year multi-world governance simulation demonstrating constitutional stability.

2 The Eight Dimensions

Definition 1 (Sovereign Profile)**.** *A sovereign profile is a vector* $\mathbf{P} = (D_0, D_1, \ldots, D_7) \in [0, 1]^8$ *where each dimension* D_i *is measured from an independent source cluster. Values outside* $[0, 1]$ *are clamped; NaN and* $\pm\infty$ *are treated as 0.0.*

Table 1 presents the eight dimensions with their source clusters, measurement methods, and saturation thresholds.

2.1 Why Eight, Not Four

The original Mycelix governance credential used four abstract dimensions: identity verification (25%), reputation (25%), community trust (30%), and engagement (20%). Implementation revealed five attack vectors that the 4D model left unaddressed:

1. **The Confident Liar**: Strong identity, good reputation, high trust, active engagement—but systematically spreading misinformation. D_0 (Epistemic Integrity) closes this gap.
2. **The Digital Ghost**: Entirely digital presence with no physical footprint. D_1 (Thermodynamic Yield) and D_2 (Network Resilience) require interaction with physical reality.
3. **The Hoarder**: Accumulates resources without reciprocating. D_3 (Economic Velocity) measures circulation, not accumulation.
4. **The Credential Fossilizer**: Historical qualifications without current competence. D_7 (Domain Competence) with Ebbinghaus decay ensures only living skills count.
5. **The Value Drifter**: Contributions diverging from community consensus. D_6 (Semantic Resonance) makes alignment measurable and continuous.

2.2 Dimensional Independence

Each dimension is measured from a *different* Holochain DNA cluster. The knowledge cluster cannot influence energy records; the governance cluster cannot fabricate craft credentials. This architectural independence is the primary Sybil defense: compromising one cluster provides no advantage in other dimensions.

The normalization formula for each dimension is:

$$D_i = \min\left(1, \frac{\text{count}_i}{\text{baseline}_i}\right) \times q_i \times w_{\text{recency},i} \quad (1)$$

where baseline_i is the saturation threshold, q_i is a quality multiplier, and $w_{\text{recency},i}$ is a recency weight that downweights stale activity.

Table 1: The eight dimensions of the sovereign profile. Each dimension is measured from an independent source cluster and saturates at a defined activity level.

Dim	Name	Source Cluster	Measures	Saturation
D_0	Epistemic Integrity	knowledge/claims	Truth-telling record	50 validated claims
D_1	Thermodynamic Yield	energy/grid	Physical energy contribution	30 verified records
D_2	Network Resilience	commons/mesh-time	Node uptime, bandwidth	720 time anchors
D_3	Economic Velocity	finance/TEND	Anti-hoarding compliance	50 TEND exchanges
D_4	Civic Participation	governance/voting	Jury duty, voting, proposals	20 governance actions
D_5	Stewardship & Care	attribution/reciprocity	Verified commons labor	30 pledges
D_6	Semantic Resonance	core-FL/hyperfeel	Community value alignment	FL Hamming similarity
D_7	Domain Competence	craft/credentials	Peer-verified expertise	3 living credentials

3 Constitutional Envelope

The sovereign profile operates within a *constitutional envelope*—invariants enforced at the DHT integrity validation layer, below any coordinator zome, below any governance vote.

3.1 Hardcoded Invariants

Sixteen parameters are compiled into WASM and enforced at validation:

Table 2: Constitutional invariants (non-exhaustive). All enforced at DHT validation layer.

Invariant	Value	Purpose
Max dimension weight	0.50	Prevent single-axis capture
$\lambda_{\min}$	0.001/day	Protect long-term stewards
$\lambda_{\max}$	0.020/day	Ensure only actives govern
Min maturation	72 hours	Prevent instant bootstrap
Grace period	30 days	Prevent surprise demotion
Ramp period	30 days	Prevent sudden parameter changes
AI max tier	Steward	Human sovereignty
Membership term	365 days	Prevent permanent tenure
Veto override	67%	Break Guardian tyranny
Override window	48 hours	Community mobilization

3.2 Seven Unamendable Rights

Seven invariants form an *unamendable core*—even a 100% unanimous vote cannot remove them:

1. Veto override mechanism (67% supermajority)
2. Consciousness gating (governance via sovereign profile)
3. Term limits (365-day maximum without renewal)
4. Emergency power limits (3 sessions, 14 days each, 30-day cooldown)
5. Permission-less enforcement (any Observer can trigger expiration)
6. Fork rights (any subset can migrate)
7. Right to exit (withdrawal without penalty)

3.3 Mandatory Decay

All dimensions decay exponentially without continued engagement:

$$S_{\text{decayed}} = S_{\text{raw}} \cdot e^{-\lambda \cdot \Delta t} \tag{2}$$

where λ is community-configurable within $[\lambda_{\min}, \lambda_{\max}]$. $\lambda = 0$ is constitutionally impossible, preventing permanent oligarchy. Changes to λ ramp linearly over a minimum of 30 days to prevent sudden disenfranchisement.

Proposition 1 (No Permanent Power)**.** *For any $\lambda > 0$ and any initial score $S_0 \in (0, 1]$, there exists a finite time T^* such that $S(T^*) < \theta_{Observer}$. No participant retains governance power indefinitely without continuous engagement.*

4 Privacy Architecture

The sovereign profile never leaves the participant's device. Governance proofs use

zero-knowledge STARK proofs (Winterfell 0.13.1 [Winterfell, 2023]):

Definition 2 (Sovereign Threshold Proof)**.** *Given a public threshold θ and a public commitment $c = SHA\text{-}256(\texttt{SOVEREIGN:v1:}did : score : \lambda : elapsed)$, prove that $S_{decayed} \geq \theta$ without revealing S_{raw}, individual dimensions, λ, or Δt.*

Table 3: ZKP performance (release build, 32-row range proof).

Metric	Value
Prove time	∼1 ms
Verify time	∼0.3 ms
Proof size	∼7.5 KB
Public inputs	threshold, commitment
Private inputs	decayed_score, agent_did

The commitment hash prevents replay attacks: a proof generated for one score/time combination cannot be reused for another.

5 Hyperdimensional Encoding

For efficient similarity search and tier derivation, profiles are encoded as 16,384-bit binary hypervectors (BinaryHV):

1. Each dimension owns a weighted fraction of the 16,384 bits, allocated according to the community's `DimensionWeights`.
2. Active bits within each dimension's region are proportional to the dimension value: $\text{bits}_i = \lfloor w_i \times 16384 \times D_i \rfloor$.
3. The encoding guarantees **monotonicity** (higher values produce strictly more bits) and **determinism** (same profile + weights = same HV).

Tier derivation uses popcount: tier $= f(\text{popcount(HV)})$ where f maps to calibrated thresholds.

This encoding enables efficient operations:

- **Profile similarity**: Hamming distance between two profiles in $O(d/64)$ via POPCNT.
- **Community consensus**: Majority-vote bundle of profiles produces the "average citizen" HV.
- **Anomaly detection**: Profiles far from the consensus HV trigger investigation.

6 Security Analysis

6.1 Sybil Cost Analysis

Table 4: Cost of Sybil attack by credential dimensionality.

System	Dims	Sybil Cost
Token voting	1	Cost of tokens (flash-loanable)
Proof of Humanity	1	Cost of biometric fraud
Gitcoin Passport	∼3	Cost of social account farming
4D Consciousness	4	Months of multi-dimensional en
8D Sovereign	**8**	**Months across 8 independen**

The key insight is that dimensions measured from *independent source clusters* compound the attack cost multiplicatively, not additively. Compromising the knowledge cluster provides zero advantage in the energy cluster.

6.2 Dimensional Capture Prevention

The constitutional constraint $\max(w_i) \leq 0.50$ ensures that no single dimension can become the sole gateway to power. Even if a community attempts to set $w_0 = 1.0$ (making epistemic integrity the only dimension that matters), the DHT validation layer rejects the configuration.

6.3 Temporal Attack Resistance

Mandatory decay ($\lambda > 0$) prevents two temporal attacks:

1. **Credential fossilization**: Historical qualifications cannot provide permanent governance power.
2. **Flash participation**: The 72-hour maturation period prevents instant credential bootstrapping.

7 Evaluation

7.1 Unit Tests

The `sovereign-profile` crate contains 87 unit tests across five modules:

- `lib.rs`: 29 tests (core types, scoring, tier derivation)
- `decay.rs`: 28 tests (exponential decay, constitutional bounds, ramp transitions)
- `hdc.rs`: 17 tests (HDC encoding, monotonicity, determinism)
- `collectors.rs`: 12 tests (normalization, saturation, recency)
- `compat.rs`: 1 test (4D→8D conversion)

7.2 Migration

The 8D system was deployed across the full Mycelix platform in a single migration:

- 74 zome coordinators across 8 clusters migrated from `gate_consciousness()` to `gate_civic()`
- Zero regressions in 9,600+ existing tests
- Backward-compatible: legacy 4D credentials auto-convert to 8D

7.3 300-Year Governance Simulation

A multi-world simulation (5 seeds, 50 agents per world, 10,000 cycles per world) tested the constitutional invariants over 300 simulated years:

- **Zero Guardian vetoes attempted** across all seeds (deterrence effect)
- **100% community survival** (no governance collapse)
- **Hostile Guardian scenario**: 84 veto attempts over 50 years, 100% community override success rate
- **Governance authority evolution**: Mission Control → Local Sovereignty → Federation → Confederation

The zero-veto result demonstrates the strongest form of constitutional success: the constraint prevents the behavior it targets without ever being invoked.

8 Related Work

Proof of Humanity [Kleros, 2021] verifies biological uniqueness via video attestation but provides no governance engagement signal. **Worldcoin** [Worldcoin, 2023] uses iris biometrics for global uniqueness but raises privacy concerns and measures nothing about trustworthiness. **Gitcoin Passport** [Gitcoin, 2023] aggregates multiple identity signals but remains identity-focused rather than governance-focused. **BrightID** [BrightID, 2020] uses social graph analysis for Sybil resistance but does not produce governance credentials. **Soulbound Tokens** [Weyl et al., 2022] propose non-transferable attestations but lack temporal dynamics (no decay, no constitutional bounds). **Verifiable Credentials** [W3C, 2022] provide a data model for claims but do not specify governance semantics.

Our approach differs in three ways: (1) eight independent dimensions measured from physical sources, not abstract identity signals; (2) mandatory temporal decay preventing credential fossilization; (3) constitutional invariants enforced at the validation layer, not the application layer.

9 Conclusion

The 8-Dimensional Sovereign Profile introduces a new primitive for decentralized governance: a multi-dimensional, temporally-decaying, privacy-preserving civic credential computed from independent physical sources. The constitutional envelope—hardcoded invariants below any governance vote—provides the strongest available form of pre-commitment against dimensional capture, permanent oligarchy, and Guardian tyranny.

The system is deployed across 141+ Holochain zomes, validated through 300-year simulations, and released as an open-source Rust crate (`sovereign-profile`, AGPL-3.0) with 87 unit tests and zero HDK dependency. We believe this architecture offers a structural alternative to both capital-weighted governance (plutocracy) and uniform governance (ochlocracy): governance power that is earned, multi-dimensional,

temporally bounded, and constitutionally constrained.

Availability. Code: https://github.com/Luminous-Dynamics/mycelix (AGPL-3.0). The `sovereign-profile` crate is published on crates.io.

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